

1] What are control structures in PL/SQL? Explain the IF-THEN and LOOP control structures.

Ans:

Control Structures in PL/SQL

Control structures in PL/SQL allow you to control the flow of execution in your program. These structures include conditional statements and looping mechanisms that help in decision-making and repeating tasks. They help in implementing business logic and handling different scenarios within a PL/SQL block.

Types of Control Structures in PL/SQL:

Conditional Statements: These are used to execute certain code only if a condition is true.

IF-THEN

IF-THEN-ELSE

IF-THEN-ELSIF-ELSE

Loops: These are used to repeat a block of code multiple times.

LOOP

WHILE LOOP

FOR LOOP

1. IF-THEN Control Structure:

The IF-THEN structure is used to execute a block of code only if a specified condition is true. It is the simplest form of conditional logic in PL/SQL.

Syntax:

IF condition THEN

```
-- Code to execute if the condition is true  
END IF;
```

2. LOOP Control Structure:

The LOOP structure is used to repeat a block of code an indefinite number of times. It continues to loop until it is explicitly terminated using a EXIT statement or when a specific condition is met.

Syntax:

LOOP

```
-- Code to execute repeatedly
```

```
EXIT WHEN condition; -- Exit condition to stop the loop
```

```
END LOOP;
```